

Starter!

Answer the 3 questions on your own and then check with a partner.
Be ready to share your answers with the class.

1. What does it mean if a substance is pure?
2. Which of these items are pure substances?



Sodium chloride ; oxygen ; sea water ; gold ; orange juice ; copper oxide ;

Silver nitrate ; soil ; black ink ; potassium

3. Of the items in **question 2**, which are elements?

Purity

Lesson objectives

1. What is meant by purity?
2. How can we calculate percentage purity?
3. How difficult is it to get a pure product?

Success criteria

Lesson Title	Working towards	Working at	Working beyond
The structure of the atom	I can list the sub-atomic particles found in an atom.	I can describe the structure of an atom.	I can describe discoveries that led to the atomic model we use today.
Purity	I can describe what an element is.	I can explain what purity means.	I can calculate and interpret percentage purity.
Filtration	I can follow steps to filter a mixture.	I can describe the purpose and parts of the filtration setup.	I can explain why filtration separates substances using particle size.
Simple Distillation	I can describe what distillation does.	I can label a simple distillation setup.	I can explain how boiling and condensation separate substances.
Crystallisation	I can describe how crystals can form.	I can carry out a crystallisation method safely.	I can explain how solubility and evaporation lead to crystal formation.
Chromatography	I can state that coloured substances can be separated using chromatography.	I can describe and carry out a chromatography experiment.	I can analyse and explain chromatograms to identify substances.



**Do you know the differences
between these 4 gold jewelry?**

Pure elements

Pure gold = 24 carat

If its less than 24 carat, it is an alloy.

This is a measure of its purity.



24K Gold

Pure elements

The more gold it has, the higher its purity.

18 carat gold has 18 parts out of 24 that are gold. (the rest are made out of other metals, such as silver or copper).

18 carat gold has a purity of 75%.

To calculate this: $\frac{18}{24} \times 100 = 75\%$



24K Gold

Pure elements

The photograph shows samples of 8 carat, 14 carat, 18 carat, and 24 carat gold. You can see that the colour changes from slightly coppery to yellow-gold.



	24-CARAT	22-CARAT	18-CARAT	14-CARAT
PURITY				
PART GOLD (%)	 99.9%	 91.6%	 75%	 58.3%
KEY USES	 INVESTMENT BARS AND COINS	 COINS AND PLAIN JEWELLERY	 FINE JEWELLERY	 COMMON JEWELLRY



Pure elements

Silver are sold with number marked
Number represents parts per thousand
that are silver.

So, silver marked 925, have 925 silver
atoms out of 1000, and 75 atoms of
some other metal.

Calculation: $\frac{925}{1000} \times 100 = 92.5\%$



So it is 92.5% pure silver



**Is made up of
Carbon element**

What are some properties of pure diamond?

Colourless

Hard

Translucent (lets light pass through)

Shiny



A pure diamond

If few carbon atoms per million are replaced with **Nitrogen**



A yellow diamond

If some carbon atoms are replaced by atoms of element, **Boron**.



A blue diamond

Rarest of all!

One atom per 1000 of
carbon atoms is replaced
by **nitrogen, nickel /
hydrogen**



A green diamond

Questions

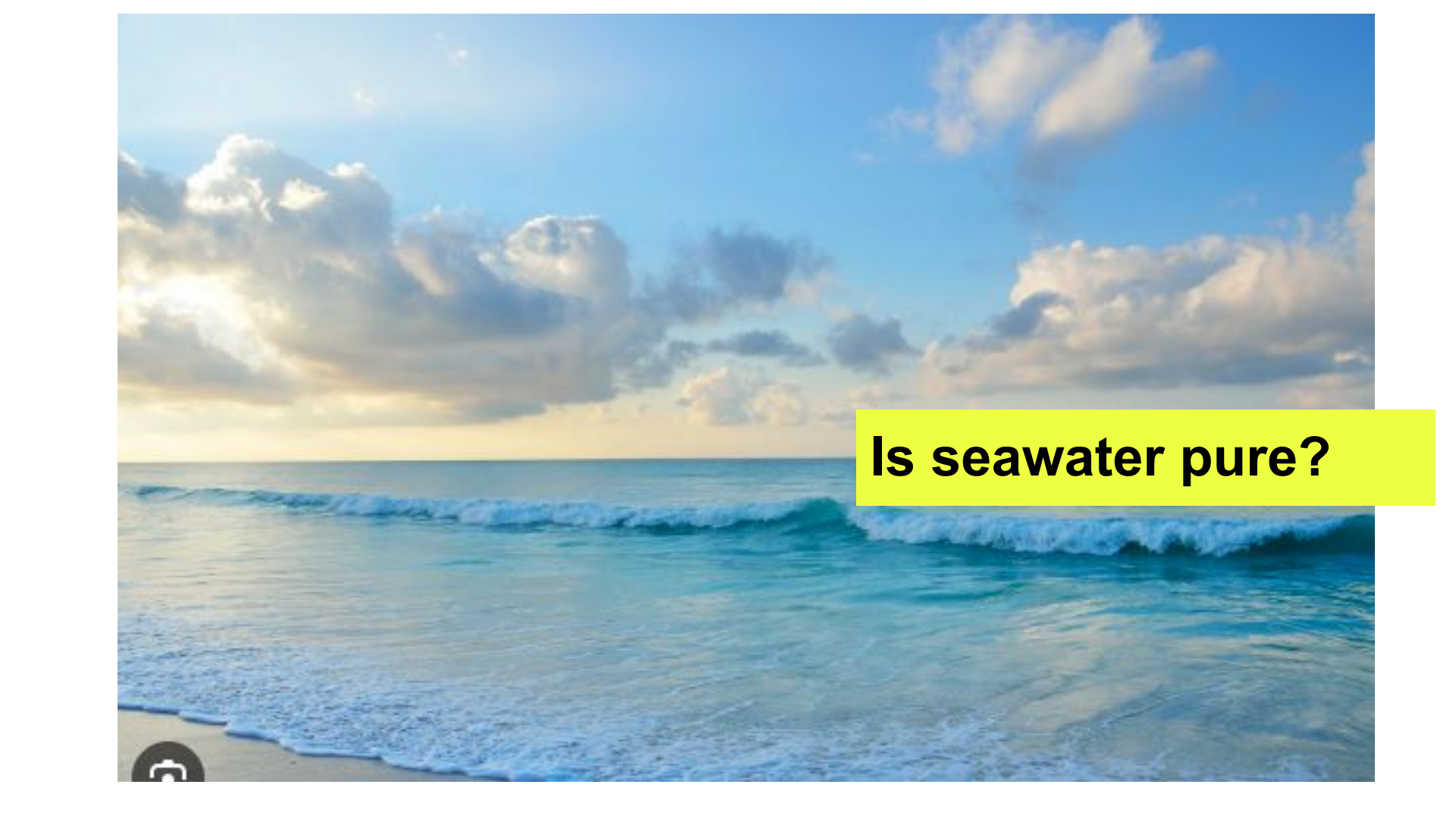


10:00

1. What percentage of 9 carat gold is gold?
2. What percentage of silver is in silver marked 900?
3. Which element mixed with carbon in diamonds make them blue?
4. Which elements may cause a diamond to be green?

Questions

1. What percentage of 9 carat gold is gold? **37.5%**
2. What percentage of silver is in silver marked 900? **90%**
3. Which element mixed with carbon in diamonds make them blue? **Boron**
4. Which elements may cause a diamond to be green?
Nitrogen, nickel / hydrogen



Is seawater pure?

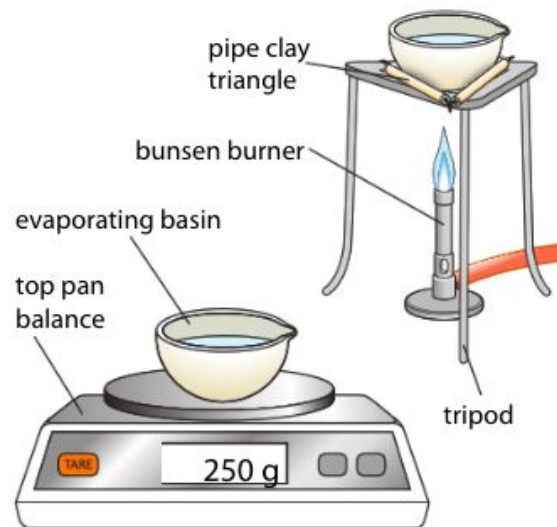
Finding the mass of salts in seawater

You will need:

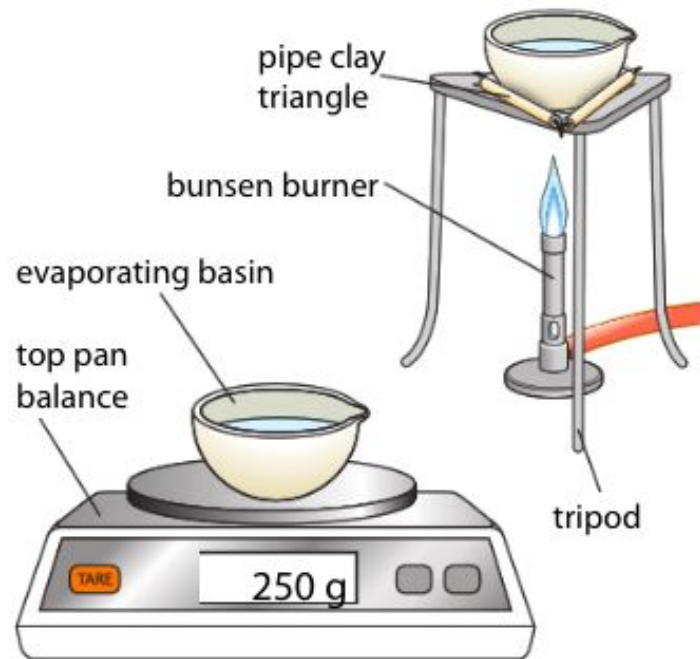
- evaporating basin
- tongs
- Bunsen burner
- tripod
- pipe-clay triangle
- safety glasses
- top pan balance
- seawater sample

Method

- 1 Read through the method and think very carefully about any risks there may be when you carry out this task. Write a risk assessment.
- 2 Place an evaporating basin on a top pan balance and record its mass.
- 3 Add some seawater and find the mass of the seawater.
- 4 Heat the seawater until it begins to spit.
Remove from the heat and allow the rest of the water to evaporate.
- 5 When there is no longer any water remaining and the basin is cool, find the mass of the salts.



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10:00

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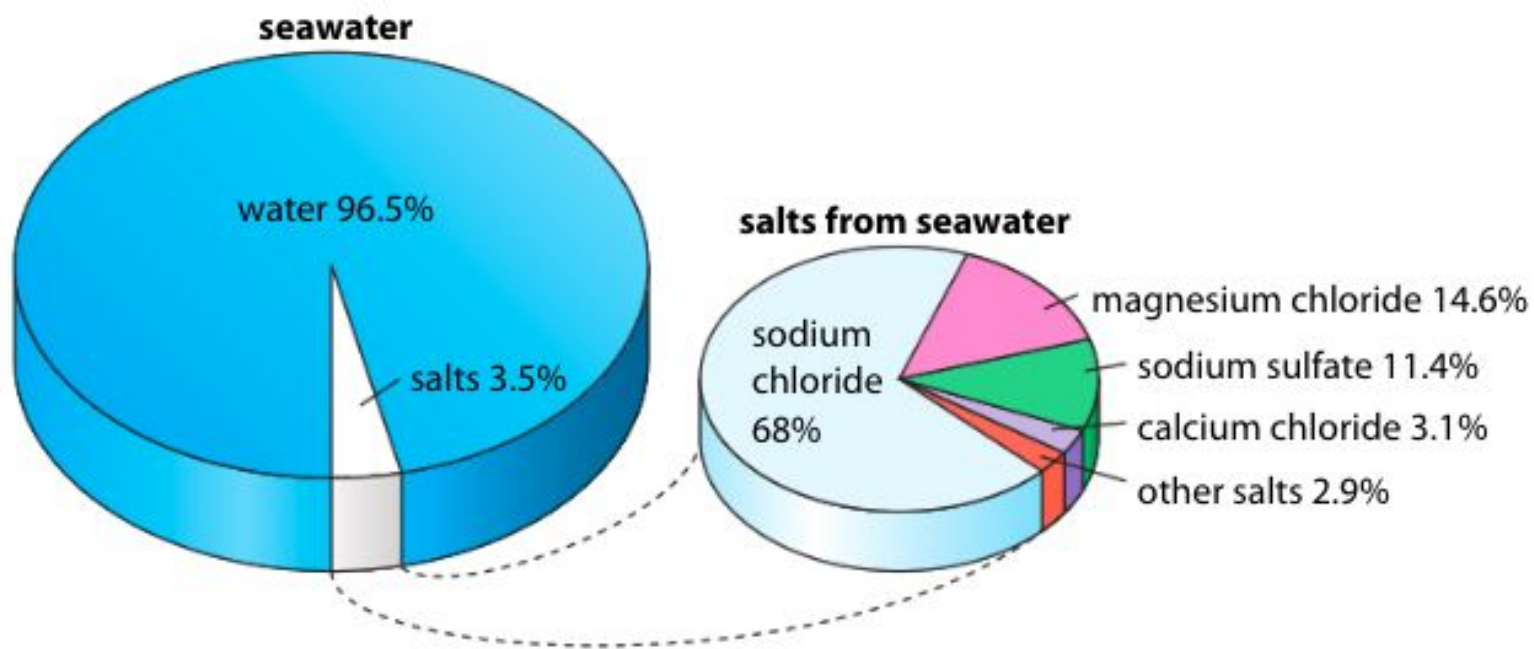
- 1 What mass of salts did you obtain from the seawater?
- 2 What percentage of the seawater is this?
- 3 About 68% of the salt in seawater is sodium chloride. Estimate the mass of sodium chloride in your sample.
- 4 Is this about what you expected? If not, why not?
- 5 What difficulties did you have carrying out this investigation? How did you try to overcome them?
- 6 What safety measures did you have in place whilst carrying out this investigation?



Questions

10:00

- 5 Draw up a table to show the percentage of salts found in seawater.
- 6 What mass of magnesium chloride would you expect to find in the seawater sample?



The salt that is obtained from this seawater is only 68% sodium chloride.
The mass of sodium chloride in 1000 g seawater is: